

Brief Review of Logistic Regression

- **Estimation**

Learning Objectives

- Build the logistic regression model
- Estimate the parameters in the logistic regression model

Classification Methods

Data: $(Y_i, x_{i1}, \dots, x_{i,p-1})$ for $i = 1, \dots, n$, where Y_i is the class label.

Two **probability-based** approaches to develop classifiers:

- Model the conditional densities $f_k = p(X|Y = k)$ at a given k -th class
 - **Normality Assumption on X**: LDA, QDA, Naïve Bayes
- Model the conditional density $P(Y = k | X)$ directly:
 - **Bernoulli assumption on Y**: logistic regression

Basic Concepts

- **Probability**

- The probability that an individual is in a specific category

- The probability of a “Yes” event is $P(Y_i = 1) = \pi_i$

- **Odds**

- The odds (or ratio) of observing an event vs. not observing that same event. $Odd_i = \frac{\pi_i}{1-\pi_i}$

- **Odds Ratio**

- The ratio of odds between two odds for two probabilities

$$\theta = \frac{Odd_1}{Odd_2} = \frac{\pi_1 / (1 - \pi_1)}{\pi_2 / (1 - \pi_2)}$$

(Binary) Logistic Regression

- **Data**: $(Y_i, x_{i1}, \dots, x_{i,p-1})$ for $i = 1, \dots, n$, where $Y_i \in \{0, 1\}$ is the class label.

- **Logistic Regression Model** is defined as two components:

- Model the response Y as Bernoulli distribution:

$$P(Y_i = 1) = \pi_i \text{ and } P(Y_i = 0) = 1 - \pi_i$$

- Link the model parameters to the independent X variables:

$$\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{i,p-1}$$

[Here p = number of β coefficients]

Conditional Probability

- Under the **Logistic Regression model**, at a given $X = (x_1, \dots, x_{p-1})$, the conditional probabilities are

$$P(Y = 1|X) = \pi = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{p-1} x_{p-1}}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{p-1} x_{p-1}}};$$

and

$$P(Y = 0|X) = 1 - \pi = \frac{1}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{p-1} x_{p-1}}}$$

Statistical Inference

Statistical Questions in Logistic Regression:

- How to estimate the β parameters in the logistic regression model from the training data?
- How to conduct hypothesis testing or confidence interval?
- How to use the logistic regression model for prediction?

Maximum Likelihood Estimation

- The likelihood function of the logistic regression model is

$$\begin{aligned} L(\boldsymbol{\beta}) &= \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \\ &= \prod_{i=1}^n \left(\frac{e^{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{ip-1}}}{1 + e^{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{ip-1}}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{ip-1}}} \right)^{1-y_i} \\ &= \prod_{i=1}^n \frac{e^{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{ip-1})}}{1 + e^{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{ip-1}}} \end{aligned}$$

- The **maximum likelihood estimator** (MLE) $\hat{\boldsymbol{\beta}}$ of the $\boldsymbol{\beta}_i$'s can be found by maximizing $L(\boldsymbol{\beta})$.

Asymptotic Properties of MLE

- The maximum likelihood estimator (MLE) $\hat{\beta}$ has nice asymptotic properties:

$$\hat{\beta} \sim N(\beta, I_{p \times p}^{-1})$$

where $I_{p \times p}$ is the observed Fisher Information Matrix that is defined by the **negative** value of the **second-order derivatives** of the log-likelihood function ($\log L(\beta)$) at $\hat{\beta}$, i.e.,

$$I_{p \times p} = \left(- \frac{\partial^2 \log L}{\partial \beta_i \partial \beta_j} \right) \Big|_{\hat{\beta}}$$

Logistic regression as a Latent variable model

Data: $(Y_i, x_{i1}, \dots, x_{i,p-1})$ for $i = 1, \dots, n$, where $Y_i \in \{0, 1\}$ is the class label.

Logistic regression as a Latent Variable Model:

1. There is an unobserved or latent continuous variable

$$Y_i^* = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{i,p-1} + \epsilon_i$$

2. We only observe the dichotomous response $Y_i = \begin{cases} 0, & \text{if } Y_i^* < 0 \\ 1, & \text{if } Y_i^* \geq 0 \end{cases}$

3. The error terms ϵ_i are iid with a **logistic distribution**, i.e.,

with pdf $f(\epsilon) = \frac{\exp(-\epsilon)}{[1+\exp(-\epsilon)]^2}$ or cdf $F(\epsilon) = \frac{1}{1+\exp(-\epsilon)}$

Other Link Function

- **Data**: $(Y_i, x_{i1}, \dots, x_{i,p-1})$ for $i = 1, \dots, n$, where $Y_i \in \{0, 1\}$ is the class label.
- **Generalized Linear Model**: Two steps
 - $P(Y_i = 1) = \pi_i, \quad P(Y_i = 0) = 1 - \pi_i$
 - $g(\pi_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{i,p-1}$where $g(\cdot): (0,1) \rightarrow (-\infty, \infty)$ is called a **link function**.
- Other link functions are available:
 - **Normit/Probit Link**: $g = \Phi^{-1}$, where $\Phi(t) = P(N(0,1) \leq t)$ is the cdf of the normal distribution. E.g., **the error terms ϵ_i are iid with $N(0, \sigma_\epsilon^2)$** .

Summary

- Logistic Regression Model
- The maximum likelihood estimation (MLE) in logistic regression
- The asymptotic properties of MLE
- A comparison and logistic regression and LDA